

The Hyperoctahedral Group on the Binary Cube Fixed Points, the Full σ -MGF, and Exact Verification

Executable companion note

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Abstract

Let $B_n = C_2^n \rtimes S_n$ act affinely on $X_n = \mathbb{F}_2^n$ and let $V_n = \mathbb{C}[X_n]$. We prove the proposed fixed-point distribution and moment formulas, derive the exact signed-cycle-type expression for the full σ -MGF, and describe an independent exhaustive verification. The program constructs every $2^n \times 2^n$ permutation matrix, computes the target directly, traverses the relevant monomial-basis orbits, and compares both results with the signed-cycle formula for $1 \leq n \leq 5$. All checks pass; the largest case has $|B_5| = 3840$ matrices of size 32×32 .

1 The action and the target

Write elements of B_n as $g = (b, \pi)$, where $b \in \mathbb{F}_2^n$ and $\pi \in S_n$, and set

$$g \cdot x = b + \pi x, \quad x \in X_n.$$

On the basis $\{e_x : x \in X_n\}$ of V_n , the corresponding matrix is

$$P_g e_x = e_{g \cdot x}. \tag{1}$$

Thus P_g is a $2^n \times 2^n$ permutation matrix. If $c_d(g)$ denotes the number of d -cycles of g on the cube vertices, the generalized permutation Molien identity gives

$$\mathcal{H}_n(\mathbf{t}) = \mathbb{E}_{g \in B_n} \prod_{d \geq 1} Q_d(\mathbf{t})^{c_d(g)}, \quad Q_d(\mathbf{t}) = \prod_{a \geq 1} (1 - t_a^d)^{-1}. \tag{2}$$

Indeed, a d -cycle permutation block has determinant $\det(I - tP_d) = 1 - t^d$. Equivalently,

$$\mathcal{H}_n(\mathbf{t}) = \mathbb{E}_g \prod_{a \geq 1} \det(I - t_a P_g)^{-1},$$

which is the intended Exp_σ expression.

For a partition $\tau = (\tau_1, \dots, \tau_s)$, its monomial coefficient has the useful orbit interpretation

$$[m_\tau] \mathcal{H}_n = \dim \left(\bigotimes_{j=1}^s \text{Sym}^{\tau_j}(V_n) \right)^{B_n} = \# \left(B_n \setminus \prod_{j=1}^s \text{Sym}^{\tau_j}(X_n) \right). \tag{3}$$

Here $\text{Sym}^r(X_n)$ means the set of r -multisets of vertices. This supplies a direct calculation independent of determinant algebra.

2 The fixed-point law

Define

$$F_n(g) = \text{Tr}(P_g) = \# \text{Fix}_{X_n}(g).$$

Theorem 1 (Exact conditional law). *For uniform $b \in \mathbb{F}_2^n$ and fixed $\pi \in S_n$, let $C(\pi)$ be the number of coordinate cycles of π . Then*

$$F_n(b, \pi) = \begin{cases} 2^{C(\pi)}, & \text{with probability } 2^{-C(\pi)}, \\ 0, & \text{with probability } 1 - 2^{-C(\pi)}. \end{cases}$$

Proof. On a coordinate cycle $(i_1, \dots, i_\ell)$, the fixed equation $x = b + \pi x$ becomes a cyclic system

$$x_{i_1} = b_{i_1} + x_{i_\ell}, \quad x_{i_2} = b_{i_2} + x_{i_1}, \quad \dots \quad x_{i_\ell} = b_{i_\ell} + x_{i_{\ell-1}}.$$

Adding in $\mathbb{F}_2$ shows that it is solvable only if $\sum_{j=1}^\ell b_{i_j} = 0$. This condition is also sufficient: after one coordinate is chosen, the remaining coordinates are forced. Consequently each coordinate cycle imposes one independent parity condition on b and, when solvable, contributes two solutions. There are $C(\pi)$ independent conditions and $2^{C(\pi)}$ solutions. $\square$

Let $\begin{bmatrix} n \\ k \end{bmatrix}$ be the unsigned Stirling number of the first kind. Since there are $\begin{bmatrix} n \\ k \end{bmatrix}$ permutations with k cycles, Theorem 1 yields

$$\mathbb{P}(F_n = 2^k) = \frac{\begin{bmatrix} n \\ k \end{bmatrix}}{n! 2^k}, \quad 1 \leq k \leq n. \quad (4)$$

The remaining probability is at zero. Using $\sum_k \begin{bmatrix} n \\ k \end{bmatrix} x^k = x(x+1) \cdots (x+n-1)$ gives

$$\mathbb{P}(F_n > 0) = \frac{(1/2)^{\bar{n}}}{n!} = \frac{\Gamma(n + \frac{1}{2})}{\sqrt{\pi} \Gamma(n+1)} = \frac{\binom{2n}{n}}{4^n} \sim \frac{1}{\sqrt{\pi n}}. \quad (5)$$

Hence $F_n \rightarrow 0$ in probability.

For every integer $m \geq 1$, conditioning on π instead gives

$$\mathbb{E}_b[F_n^m \mid \pi] = 2^{-C(\pi)} 2^{mC(\pi)} = 2^{(m-1)C(\pi)}.$$

Another application of the Stirling generating polynomial proves

$$\boxed{\mathbb{E}[F_n^m] = \frac{(2^{m-1})^{\bar{n}}}{n!} = \binom{n + 2^{m-1} - 1}{n}}. \quad (6)$$

In particular $\mathbb{E}F_n = 1$, $\mathbb{E}F_n^2 = n + 1$, $\mathbb{E}F_n^3 = \binom{n+3}{3}$, and $\text{Var}(F_n) = n$. Thus convergence in probability to zero coexists with divergent moments of every order $m \geq 2$.

There is also a direct Burnside proof of (6). Represent an ordered m -tuple of cube vertices by an $m \times n$ binary matrix. Coordinate permutations permute its columns, while a flip in one cube coordinate adds $\mathbf{1} = (1, \dots, 1)$ to the corresponding column. Each column is therefore remembered only as an element of $\mathbb{F}_2^m / \langle \mathbf{1} \rangle$, a set of size 2^{m-1} . The orbit is a multiset of n such column types, so

$$\#(B_n \setminus X_n^m) = \binom{n + 2^{m-1} - 1}{n} = \mathbb{E}[F_n^m].$$

Combining this with (3) gives

$$[m_{(1^m)}] \mathcal{H}_n = \binom{n + 2^{m-1} - 1}{n}. \quad (7)$$

3 Signed cycle types and the full cycle index

A signed coordinate-cycle type is a pair of sequences $(a_\ell, b_\ell)_{\ell \geq 1}$, where a_ℓ counts length- ℓ coordinate cycles with total flip parity zero and b_ℓ counts those with total flip parity one. They satisfy

$$\sum_{\ell \geq 1} \ell(a_\ell + b_\ell) = n.$$

For a uniform element of B_n , the probability of this type is

$$w(a, b) = \prod_{\ell \geq 1} \frac{1}{(2\ell)^{a_\ell + b_\ell} a_\ell! b_\ell!}. \quad (8)$$

This follows from the usual S_n cycle-type probability and the fact that the accumulated parity on every coordinate cycle is an independent fair bit.

Let

$$R_r(a, b) = \# \text{Fix}_{X_n}(g^r)$$

for any element of signed type (a, b) . A positive length- ℓ cycle contributes $2^{\gcd(\ell, r)}$ fixed choices. For a negative cycle, write the affine map as $T(x) = Rx + e_0$, with R a cyclic rotation. If $d = \gcd(\ell, r)$, the permutation R^r has d coordinate orbits. On each orbit, the translation in T^r has parity r/d . The fixed equation is therefore solvable exactly when r/d is even; when solvable it has 2^d solutions. Independence between coordinate cycles proves

$$R_r(a, b) = \begin{cases} 2^{\sum_{\ell \geq 1} (a_\ell + b_\ell) \gcd(\ell, r)}, & \frac{r}{\gcd(\ell, r)} \text{ is even for every } \ell \text{ with } b_\ell > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

Because $R_r = \sum_{d|r} d c_d$, Möbius inversion recovers the exact vertex-cycle counts:

$$c_d(a, b) = \frac{1}{d} \sum_{r|d} \mu(d/r) R_r(a, b). \quad (10)$$

Substituting (8)–(10) into (2) proves the full finite- n formula

$$\boxed{\mathcal{H}_n(\mathbf{t}) = \sum_{\substack{(a_\ell), (b_\ell) \\ \sum_{\ell} \ell(a_\ell + b_\ell) = n}} w(a, b) \prod_{d \geq 1} Q_d(\mathbf{t})^{c_d(a, b)}}. \quad (11)$$

This is exact, not an asymptotic or a truncation.

At total degree two, the orbit interpretation gives a transparent check. The group is transitive on vertices, and pairs of vertices are classified by their Hamming distance $0, 1, \dots, n$. Therefore

$$\mathcal{H}_n = 1 + m_{(1)} + (n+1)(m_{(2)} + m_{(1,1)}) + \dots. \quad (12)$$

In particular, the coefficients do not stabilize as $n \rightarrow \infty$.

4 Independent verification strategy

The companion program avoids using the statement being tested as its only computational route.

Literal matrix computation. For every one of the $2^n n!$ pairs (b, π) , the program builds the vertex permutation and the matrix (1). It checks that the matrices are distinct permutation matrices. For a partition τ , it computes complete symmetric characters from actual matrix traces using Newton’s recurrence

$$d h_d(P_g) = \sum_{r=1}^d \text{Tr}(P_g^r) h_{d-r}(P_g), \quad h_0 = 1, \quad (13)$$

and averages $\prod_j h_{\tau_j}(P_g)$ exactly.

Direct orbit computation. The monomial basis in (3) is constructed literally. A breadth-first traversal using the n coordinate flips and $n - 1$ adjacent coordinate transpositions counts its orbits. This computation does not use traces, determinants, signed types, or the proposed formula.

Signed-type computation. All nonnegative solutions (a_ℓ, b_ℓ) of the size constraint are generated without enumerating group elements. Their rational weights (8) are checked to sum to one. Equations (9) and (10) then produce vertex cycles and the coefficient predicted by (11).

The tests also compare the entire convergent function, not only a finite Taylor truncation. At $(t_1, t_2) = (0.037, 0.061)$ they evaluate

$$\frac{1}{|B_n|} \sum_g \frac{1}{\det(I - t_1 P_g) \det(I - t_2 P_g)}$$

by dense determinants, by actual vertex cycles, and by the signed-type sum.

5 Exact results

The following coefficient vectors are ordered as

$$\tau = (1), (2), (1, 1), (3), (2, 1), (1, 1, 1).$$

Every entry agrees across literal matrices, direct orbit traversal, and the signed-type formula.

n	$ B_n $	$\dim V_n$	$\mathbb{P}(F_n > 0)$	common coefficient vector
1	2	2	1/2	(1, 2, 2, 2, 3, 4)
2	8	4	3/8	(1, 3, 3, 4, 7, 10)
3	48	8	5/16	(1, 4, 4, 7, 13, 20)
4	384	16	35/128	(1, 5, 5, 11, 22, 35)
5	3840	32	63/256	(1, 6, 6, 16, 34, 56)

For $n = 1, \dots, 5$, the program additionally verifies, element by element, the conditional parity law and reconstructs every vertex-cycle count from (9)–(10). It checks moments $m = 1, 2, 3, 4$, giving 20 exact moment comparisons, and the six displayed coefficients, giving 30 three-way coefficient comparisons. In the largest numerical case, the matrix determinant average is

$$1.1620117926716296,$$

and its maximum discrepancy from the cycle and signed-type evaluations is 1.74×10^{-14} , consistent with floating-point roundoff. All integer and rational comparisons are exact.

The range $n \leq 5$ is a practical exhaustive choice: it includes all 3840 elements at $n = 5$ and orbit spaces as large as $|X_5|^3 = 32768$, while the group order grows as $2^n n!$. The proof and signed-type algorithm remain valid for every n ; only brute-force matrix enumeration is deliberately bounded.

6 Reproduction

From the repository root, run

```
marimo run for_this_guy/finite_group_sigma_mgf/  
hyperoctahedral_cube/notebook.py
```

The notebook allows inspection of individual matrices and signed types and shows every comparison table. The noninteractive regression test is

```
pytest -q tests/test_hyperoctahedral_cube.py
```

7 Conclusion

The proposed formulas are correct. A random cube symmetry has no fixed vertex with probability tending to one, but its rare nonzero fixed sets are large enough that $\mathbb{E}F_n = 1$ and all higher moments diverge. The signed-cycle power formula determines every vertex cycle and hence the complete σ -MGF through (11). Exhaustive matrix, orbit, and signed-type computations agree throughout the representative range $1 \leq n \leq 5$.